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Task Context Shapes Component Roles in Language Models

The Same Attention Head Does Different Jobs Depending on the Task— and Response Maps Predict This Redistribution Across Five Model Families

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The same attention head can carry out fundamentally different computations depending on the task context, even when the factual content of the prompt is held constant. Across five model families (1.5–14B), switching tasks redistributes influence among heads, and a linear *response map* fitted per task predicts a downstream head's change from an upstream activation edit, cutting mean squared error by 53.6% in Gemma-2-9B.

AT A GLANCE

Problem: Mechanistic interpretability studies heads in a single-task context; it is unknown whether those functional roles transfer.

Why it matters: Safety interventions derived from single-task circuits may fail or backfire under different task contexts.

Method: Graph-fact prompts isolate task context from factual content; activation patching measures cross-task influence redistribution.

Result: Mean cosine similarity between influence profiles across tasks is 0.41; response maps transfer with 50–70% of MSE reduction preserved.
Five models, five task types, causal intervention validation.

The Big Picture

Mechanistic interpretability has made progress identifying attention heads responsible for specific behaviors—induction heads for copying, name-mover heads for indirect object identification. But almost all such analyses are conducted under a *single* task context.

Mohapatra and Bassett ask: does the same head remain responsible for the same computation when the task changes, even if the factual content of the prompt is unchanged? Using carefully controlled **graph-fact** prompts that hold syntactic structure and factual entities constant while varying only the task instruction, they find that component roles are substantially reshuffled.

This challenges the view of neural circuits as fixed functional units and suggests that safety interventions must account for task-dependent rerouting.

Why Existing Approaches Fall Short

Single-task circuit tracing (e.g., path patching, causal tracing) identifies the heads responsible for a behavior in one setting but provides no prediction of how those heads behave under a different task with identical factual content.

Universal probes train on a fixed dataset to locate where facts are stored. They miss the dynamic routing that occurs at task-switch time.

Attention pattern visualization shows where a head attends but not how its output influences downstream layers. Influence requires causal intervention, not observation.

Core Mechanism: Graph Facts and Response Maps

Graph-fact pairs are minimal-pair prompts that share a factual entity (e.g., a country capital) at the same token position across tasks but differ only in the task instruction (factual recall, translation, analogy, completion, QA). This controls for syntactic and positional confounders.

Influence measurement. For each head (h, ℓ), the indirect effect under task t is:

$$\mathrm{IE}^{(t)}(h,\ell) = \frac{1}{N}\sum_{i=1}^{N}\Big[p_\theta(a_i^{*(t)}|x_i^{(t)}, \mathrm{patch}_{h,\ell}) - p_\theta(a_i^{*(t)}|x_{\mathrm{corrupt},i}^{(t)})\Big]$$

Response maps. A linear map $R^{(t)}_{(h,\ell)\to(h',\ell')}$ predicts the output change at downstream head (h', ℓ') from an activation edit at upstream head (h, ℓ) under task t:

$$\hat{\Delta}a_{h',\ell'} = R^{(t)}_{(h,\ell)\to(h',\ell')} \cdot \Delta a_{h,\ell}$$

Quality is measured by fractional MSE reduction relative to a zero predictor.

Technical Formulation

Let $\mathrm{IE}^{(t)} \in \mathbb{R}^{N_{\text{heads}}}$ be the influence vector across all heads under task t. The cross-task similarity:

$$\rho(t,t') = \frac{\mathrm{IE}^{(t)} \cdot \mathrm{IE}^{(t')}}{\|\mathrm{IE}^{(t)}\|\,\|\mathrm{IE}^{(t')}\|}$$

A value of $\rho < 1$ indicates that the relative importance of heads changes across tasks. The authors observe $\rho \in [0.18, 0.67]$ across task pairs (mean 0.41).

Response map quality:

$$\text{MSE-red} = 1 - \frac{\|\Delta a_{h',\ell'} - \hat{\Delta} a_{h',\ell'}\|^2}{\|\Delta a_{h',\ell'}\|^2}$$

Gemma-2-9B: 53.6%; Llama-3.1-8B: 41.2%; Phi-3-mini: 38.8%; Qwen3-14B: 5.0%.

Learning or Inference Procedure

1. Construct graph-fact prompts for five task types; ensure factual entities appear at the same token position.
2. For each model and task, run activation patching to compute $\text{IE}^{(t)}(h, \ell)$ over all heads.
3. Fit response maps $R^{(t)}$ on 80% of prompts; evaluate on 20%.
4. Perform targeted early-layer edits; measure downstream answer-token probability changes.
5. Compare influence profiles across tasks via cosine similarity.

What the Guarantee Says

The response maps are empirically validated: for one factual intervention, the Gemma-2-9B map reduces MSE by 53.6% vs. averaging the two task-specific maps. Cross-task transfer preserves 50–70% of this reduction, suggesting a shared sub-structure that is partially predictable from single-task analysis.

Experimental Findings

- **Models:** Phi-3-mini-3.8B, Gemma-2-9B, Llama-3.1-8B, Qwen3-14B, Mistral-7B.
- **Tasks:** Factual recall, translation, analogy, sentence completion, open-ended QA.
- Mean cross-task influence cosine similarity: **0.41** (range 0.18–0.67).
- Response map MSE reduction: Gemma-2-9B **53.6%**; Llama-3.1-8B 41.2%; Qwen3-14B 5.0%.
- Cross-task transfer: maps from factual recall transfer best to sentence completion (0.62), worst to translation (0.19).

Ablations and Interpretation

Fixed-task vs. cross-task maps: Single-task maps outperform cross-task maps by 15–20% MSE on the training task, confirming genuine task-specific routing.

Layer range: Early-layer edits (layers 3–7 of 28) are the primary routing bottleneck; mid-layer response maps are less predictive.

Attention vs. MLP: Attention heads account for 70–85% of task-dependent influence redistribution; MLP contributions are less variable across tasks.

Safety implication: A circuit marked "safe" under factual recall may be activated differently under translation or QA context. Cross-task analysis is recommended before deployment.

Reference

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